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Gary Gartling, Masayoshi Yoshimatsu, Elizabeth Hary, Renjie Bing, Vivian Didomenico, Michael J. Garabedian, Aaron M. Johnson & Ryan C. Branski

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A Preliminary Integrated Morphologic and Transcriptomic Analysis of Denervation Remodeling in Rat Laryngeal and Hindlimb Muscles

[1]Gary Gartling, PhD
[1] Masayoshi Yoshimatsu, MD, PhD
[2]Elizabeth Hary, MS
[1]Renjie Bing, MD
[1]Vivian Didomenico
[3,4]Michael J. Garabedian, PhD
[1]Aaron M. Johnson, PhD
[1]Ryan C. Branski, PhD

[1]Otolaryngology-Head and Neck Surgery, NYU Grossman School of Medicine, New York, NY

[2]Communication Science and Disorders, School of Health and Rehabilitation Sciences, University of Pittsburgh, Pittsburgh, PA

[3]Microbiology, NYU Grossman School of Medicine, New York, NY

[4]Urology, NYU Grossman School of Medicine, New York, NY

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Corresponding Author
Gary Gartling, PhD, CCC-SLP
Otolaryngology-Head and Neck Surgery
NYU Grossman School of Medicine
435 East 30th Street, 10th Floor
New York, NY 10016
gary.gartling@nyulangone.org

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Abstract

Background: The translational relevance of the rat laryngeal model has been questioned due to key interspecies differences in laryngeal muscle fiber-type and MyHC composition from humans, notably the absence of type I fibers in the thyroarytenoid muscle. This difference fueled the view that fiber-type composition alone dictates outcomes of denervation injury.

Objective: To preliminarily assess whether organ context (larynx vs. hindlimb), rather than fiber type (type I vs. type II) or subtype (IIB vs. IIA vs. IIX), influences structural and transcriptional responses to denervation and to explore the translational relevance of the rat model for future investigation.

Methods: Unilateral left recurrent laryngeal nerve and tibial nerve transections were performed in male and female Sprague Dawley rats. Five weeks following injury, muscle fiber size was quantified by minimum Feret diameter (mFD) in three laryngeal muscles (medial thyroarytenoid (MTA), lateral thyroarytenoid (LTA), and posterior cricoarytenoid (PCA) and two hindlimb muscles (gastrocnemius, soleus). Bulk RNA-seq was performed on each muscle type and pathway enrichment was used to define the molecular response.

Results: Structurally, laryngeal muscles demonstrated significantly less atrophy compared to fiber-matched hindlimb muscles. This structural resilience was mirrored at the molecular level. Denervated hindlimb muscles exhibited a robust transcriptional response, characterized by pathways related to inflammation, extracellular matrix remodeling, and oxidative stress, hallmarks of a classic injury response. In contrast, an attenuated and selective transcriptional response was observed in the laryngeal

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muscles: the TP53 stress-response network was activated in the PCA, modest changes in mitochondrial pathways were observed in the MTA, and no significantly enriched pathways were observed in the LTA. Within the larynx itself, type IIB fibers were more susceptible to atrophy than putative type IIX/IIX-like fibers, suggesting an alternative fiber-subtype influence.

Conclusions: These preliminary findings suggest organ context is a primary determinant of denervation-associated remodeling, with fiber subtype serving as a secondary modifier. Collectively, these data provide translational insight into laryngeal atrophy and support the rat larynx as a mechanistic model of organ-specific protective and remodeling responses to denervation, rather than as a strict fiber-type analog.

Keywords: Denervation; larynx; gastrocnemius; soleus; muscle atrophy; fiber type; RNA-seq; pathway enrichment; Sprague–Dawley; Rat; Recurrent Laryngeal Nerve

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Background

Atrophy, broadly defined as the non-specific degradation of tissue and/or cells, is a significant clinical issue affecting multiple tissues and organs. The prevalence and etiology of internal laryngeal muscle atrophy, including the thyroarytenoid (TA) and posterior cricoarytenoid (PCA) muscles, are largely unknown, but may occur due to idiopathic or iatrogenic denervation. Muscle atrophy primarily occurs through the ubiquitin–proteasome pathway and activity of E3 ligases,(1) and can be triggered by denervation arising from peripheral nerve injury, surgical nerve damage, or motor neuron disease.(2)

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The recurrent laryngeal nerve (RLN) innervates most intrinsic laryngeal muscles, including the thyroarytenoid (TA) and posterior cricoarytenoid (PCA), and is susceptible to injury during thyroidectomy, anterior cervical spine surgery, and cardiothoracic procedures because of its long anatomic course.(3–8) RLN injury remains one of the most clinically significant complications of thyroid and parathyroid surgery, with potential consequences including vocal fold paresis/paralysis, dysphonia, dysphagia/aspiration, and, in bilateral cases, airway compromise. Contemporary data indicate thyroidectomy-related RLN injury may occur through traction/stretch, separation or dissection injury, transection, compression/ligation, or thermal mechanisms, and may occur despite visual identification and intraoperative neuromonitoring.(9)

In the absence of easily accessible human donor larynges, the rat has emerged as a surrogate model for RLN injury and laryngeal muscle physiology.(10–17) However, the translational validity of the rat model is unknown due to differences in muscle fiber composition between rat and human TA muscles. Specifically, the human TA is composed of approximately 40% type I fibers and 60% type II fibers;(18) the rat TA consists entirely of type II fibers.(19) Importantly, the absence of type I fibers in the rat TA does not imply a simple or homogeneous fast-fiber phenotype. Prior studies of rat intrinsic laryngeal muscles have identified distinct fast and hybrid MyHC phenotypes, including type IIB-, type IIX-, and EO/MYH13-associated fibers, with hybrid fibers expressing more than one MyHC isoform commonly reported in TA subdivisions. The rat larynx differs from limb muscle not only in the absence of type I fibers in the TA, but also in its specialized craniofacial/laryngeal myosin repertoire.

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Fiber-type is a critical determinant of muscle metabolic properties, fatigue resistance, and regenerative capacity. Type I fibers are rich in mitochondria and support sustained aerobic activity.(20) Type II fibers primarily rely on anaerobic metabolism and are more prone to fatigue.(21) Expression of myosin heavy chain (MyHC) isoforms underlies these phenotypic differences(22) and fiber-type-specific responses to injury can differ markedly.(23) Denervation-induced atrophy typically affects both type I (slow-twitch) and type II (fast-twitch) muscle fibers. In hindlimb muscles, it is generally agreed type I fibers are more vulnerable to atrophy following denervation.(24) However, increased atrophy was reported in type II fibers compared to type I fibers in the rat hindlimb.(25) More broadly, the effects of denervation on muscle fiber size are highly context-dependent, varying with muscle type, species, time point, and experimental conditions. Type II fibers consistently undergo rapid atrophy across denervated muscles, including rat hindlimb muscles(23) and turkey leg muscles.(26) In contrast, type I fibers have diverse responses: type I fibers in the rat soleus atrophy in parallel with type II fibers**,** whereas type I fibers in the rat tibialis anterior hypertrophy during the first two months after denervation.(23) This slow-fiber hypertrophy is also seen in denervated turkey leg muscles.(26) Together, these findings suggest susceptibility to denervation-induced atrophy is difficult to generalize across muscles or species.

Despite extensive knowledge of the effects of denervation in limb musculature, the response of the internal laryngeal muscles remains poorly understood. These muscles play critical roles in vocal fold movement and airway protection, and their unique fiber-type composition and innervation suggest the potential for distinct atrophic responses. The rat larynx has emerged as a valuable experimental model to investigate

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these mechanisms. Intrinsic laryngeal muscles have unique features not typical of limb muscles. They often contain hybrid muscle fibers (fibers expressing more than one myosin type), and in some species, express superfast myosin MYH13. This MYH13 isoform, referred to as extraocular (EO), can be expressed alongside standard limb muscle isoforms. Together, these characteristics highlight intrinsic laryngeal muscles may have specialized craniofacial beyond the traditional limb muscle classification into type I and type II fibers (27,28). In rats, the lateral thyroarytenoid (LTA) frequently exhibits hybrid fast fibers (e.g., IIB ± EO/MYH13)(17) and we treat this hybrid repertoire as part of the laryngeal organ context rather than a confound to fiber-class comparisons in this preliminary analysis.

These features suggest organ context may modulate denervation outcomes beyond fiber class alone. The current study sought to elucidate how denervation differentially affects laryngeal muscle fiber types compared to limb muscles to enhance our understanding of muscle-specific atrophy and inform potential therapeutic strategies for voice and airway dysfunction following nerve injury. We examined the selective impact of denervation on muscle fiber-types and subtypes and their contribution to mechanisms of atrophy using Myosin Heavy Chain (MyHC) isoform staining to determine fiber size and RNA sequencing (RNA-seq). Regarding nomenclature, thyroarytenoid (TA) refers to the muscle complex generally and medial thyroarytenoid (MTA) and lateral thyroarytenoid (LTA) refer to the specific subdivisions. Specifically, we compared atrophy after RLN transection in medial TA (MTA) (primarily IIX),(17) LTA (enriched IIB with IIB±EO/MYH13 hybrids (17) and PCA (enriched IIB/IIA/IIX)(29) with atrophy after tibial nerve transection in soleus (predominantly I with IIA) (30,31) and

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gastrocnemius (mixed IIB/IIX with some I/IIA).(31) These muscles were selected to provide broad representation of fiber-types and to enable comparison with fast-twitch rat laryngeal muscles.

This comparison was designed to test whether organ context (larynx vs. hindlimb) or fiber-type/subtype primarily governs denervation-induced atrophy and transcriptional remodeling. We hypothesized, in this preliminary cohort, laryngeal muscles would exhibit less atrophy than fiber-matched hindlimb muscles despite shared fast (type II) subtypes with the gastrocnemius and the predominantly type I composition of the soleus. Because susceptibility of type I versus type II fibers to denervation varies across muscles and paradigms, we reasoned fiber type alone insufficiently explains vulnerability and organ-specific factors likely contribute to relative resistance of the larynx. We further hypothesized denervation would elicit muscle-specific transcriptomic programs yielding distinct patterns of differential expression and pathway enrichment in laryngeal versus hindlimb muscles. Ultimately, we sought to determine whether observed differences reflect fiber-type–specific or organ-specific mechanisms of atrophy to clarify the translational value of these preliminary data. We further sought to refine the role of the rat larynx as a mechanistic analog that captures organ-context programs relevant to human denervation despite interspecies fiber-type differences.

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Methods

All experimentation was performed according to the guidelines and regulations approved by the Institutional Animal Care and Use Committee at the NYU Grossman School of Medicine (#202400004). Animal experimentation was performed in

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accordance with ARRIVE guidelines, where applicable. Eight Sprague–Dawley rats (4 male, 4 female) were included. Animals were obtained from Charles River Laboratories (Wilmington, MA, USA), a commercial supplier. Five animals were dedicated to morphometric analysis and three to RNA-seq. To minimize animal use, hindlimb tissues from the RNA-seq cohort were also included in the morphometry, so the morphometry group sizes reflect the combined contributions of the morphometry-dedicated animals plus available tissues from the RNA-seq cohort. Each rat underwent unilateral left-sided denervation; the contralateral side served as an internal control. This paired design was selected to reduce inter-animal variability and minimize animal use. However, the contralateral side should be interpreted as an internal reference rather than an entirely uninjured biological control, because unilateral denervation may alter contralateral loading, compensatory activity, or gene expression. Rats were anesthetized using intraperitoneal injection of ketamine (80mg/kg) and xylazine (10mg/kg) and placed on 2% isoflurane and oxygen via nose cone. The neck and left gluteal region were shaved and a sterile surgical field was established with draping and three passes of alcohol and iodine. Lidocaine (7mg/kg) was injected subcutaneously at each surgical site prior to incision.

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Tibial Nerve Resection. To resect the tibial nerve, a ~3cm incision was made in the skin and underlying muscle of the gluteal region. A ~10mm segment of the tibial nerve was excised. The muscle layer was then sutured with absorbable PDS 5.0 single interrupted sutures and the skin was closed using mattress VICRYL 5.0 sutures.

Recurrent Laryngeal Nerve Resection. A ~2cm incision was made from the sternum to the hyoid region and underlying muscle to expose the larynx and RLN. A ~8mm portion

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of the left RLN was removed at the 5th tracheal ring. The muscle layer was then sutured with absorbable PDS 5.0 single interrupted sutures and the skin was closed using mattress VICRYL 5.0 sutures. The incision sites were treated with triple antibiotic ointment and each animal was given Antisedan (1mg/kg), Meloxicam (1mg/kg), and extended-release buprenorphine (0.65mg/kg), subcutaneously.

Tissue processing. Animals were euthanized five weeks post-surgery via CO_2 inhalation. This timepoint was selected based on previous data indicating pronounced TA and PCA atrophy five weeks following RLN transection.(10) The whole larynx and bilateral gastrocnemius and soleus muscles from each animal were individually embedded in optimal cutting temperature (OCT) (Sci Gen Inc, Gardena, CA, USA) compound and flash frozen. Mid portions of the whole gastrocnemius and soleus muscles were sectioned coronally at 10µm and adhered to positively charged microscope slides. The TA was sectioned coronally at the mid-membranous portion and these frozen sections were also adhered to slides. The left and right PCA muscles were individually sectioned coronally at the insertion point into the cricoid cartilage at 10µm and mounted on positively charged slides. All sections were stored at -80°C until immunolabeling.

Immunofluorescence labeling. Immunofluorescence labeling was performed using established protocols (17) to identify muscle fiber-types based on MyHC isoform expression and to delineate fiber boundaries with laminin. Frozen muscle sections dried for 10 minutes at room temperature. A hydrophobic barrier pen circled each tissue section. Sections were then incubated in 10% normal goat serum (NGS; Sigma-Aldrich, G9023; St. Louis, MO, USA) in 0.1M phosphate-buffered saline (PBS) for 90 minutes.

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Primary antibodies (DSHB, Iowa City, IA, USA) followed the canonical rodent MyHC panel: BA-F8 (IgG2b; MyHC I), SC-71 (IgG1; MyHC IIA), BF-F3 (IgM; MyHC IIB), and BF-35 (IgG1; “all-but-IIX”). Putative IIX/IIX-like fibers were operationally identified based on absence of BF-35 immunoreactivity. BF-35 labels MyHC isoforms other than IIX and recognizes type I, IIA, and IIB fibers in skeletal muscle; BF-35-negative fibers were classified as putative IIX/IIX-like fibers. Because EO/MYH13 was not directly assayed and intrinsic laryngeal muscles may contain hybrid fibers, this classification should be interpreted as an operational immunophenotype rather than definitive evidence of pure IIX MyHC expression. Consequently, IIB–EO hybrid fibers in the LTA would have been classified based on their detectable IIB epitope, and a subset of hybrid or EO/MYH13-containing fibers may be subsumed within the operational fiber categories used here. Isotype-specific secondary antibodies targeting mouse IgG2b, mouse IgG1, and mouse IgM heavy chain were used (**Table 1**).

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Table 1

Antigen	Host	Antigen Species	Isotype	Product	Dilution
Primary Antibodies					
MHC I	Mouse	Bovine	IgG2b	BA-F8 (Hybridoma)	1:50
MHC IIA	Mouse	Bovine	IgG1	SC-71 (Hybridoma)	1:100
MHC IIB	Mouse	Bovine	IgM	BF-F3 (Hybridoma)	1:100
MHC IIX (all but)	Mouse	Bovine	IgG1	BF-35 (Hybridoma)	1:100
Laminin	Rabbit	Mouse	IgG	L9393 (Sigma)	1:100
Secondary Antibodies					
Anti-MHC I	Goat	Mouse	Polyclonal	A21141 (Invitrogen)	1:100
Anti-MHC IIA	Goat	Mouse	Polyclonal	A21121 (Invitrogen)	1:100
Anti-MHC IIB	Goat	Mouse	Polyclonal	A21044 (Invitrogen)	1:100

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Anti-MHC IIX (all but)	Goat	Mouse	Polyclonal	A21121 (Invitrogen)	1:100
Anti-Laminin	Goat	Rabbit	Polyclonal	A31556 (Invitrogen)	1:100

Slides were rinsed three times for five minutes in PBS. Secondary antibodies were centrifuged for 2–3 minutes to reduce fluorescent precipitates. Hindlimb sections were incubated with secondary antibodies for 1 hour at room temperature. Laryngeal sections were incubated for 4 hours under the same conditions. Slides were rinsed in PBS and immersed briefly in Histoclear (National Diagnostics, GA, USA) to remove the hydrophobic barrier. ProLong™ Gold Antifade Mountant without DAPI (Thermo Scientific, Waltham, MA) covered the tissue sections using #1.5 rectangular coverslips (22×40 mm). Slides dried overnight at room temperature in the dark and were stored at 4°C.

Imaging. Immunofluorescent imaging was performed using the PhenoImager HT multispectral imaging system (Akoya Biosciences, Marlborough, MA, USA). Whole-slide images were acquired under consistent imaging parameters across all samples. To enable accurate segmentation of individual muscle fibers, regions of interest were magnified within QuPath software.(32) For each contralateral comparison within the same muscle fiber-type (e.g., left medial thyroarytenoid [MTA] vs. right MTA), a fixed exposure time and magnification were applied to ensure consistency. Additionally, images were captured from anatomically matched regions within each muscle (e.g., lateral head of the gastrocnemius near the muscle surface) to account for known regional differences in fiber-type distribution.(33) TIFF images were exported from QuPath for subsequent analysis.

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Image Analysis. Quantification of muscle fiber mFD was performed using SMASH (Semi-automatic Muscle Analysis using Segmentation of Histology), a MATLAB-based application (MathWorks, Natick, MA, USA).(34) TIFF images exported from QuPath were uploaded into SMASH for semi-automated segmentation. For morphometric analysis, sex distribution varied by muscle and fiber type due to differences in section quality and sample availability. Gastrocnemius putative IIX and IIB each included 2 males and 4 females; LTA IIB and MTA putative IIX each included 2 males and 3 females; soleus IIA included 2 males and 3 females; soleus MyHC I included 2 males and 2 females; and PCA IIA included 1 male and 2 females.

Because tissue quality and section availability varied across muscles and fiber types, final morphometric sample sizes differed among comparisons. Final biological replicate numbers were PCA, n = 3; soleus, n = 4–5 depending on fiber type; MTA, n = 5; LTA, n = 5; and gastrocnemius, n = 6, with sex distribution reported by muscle and fiber type in **Supplementary Table 1**.

Hindlimb samples from the RNA-seq cohort were incorporated into morphometry when possible to reduce animal use. Sex was pooled for morphometry because each denervated muscle was compared to its contralateral control, minimizing systemic sex-related baseline differences. To ensure accurate fiber detection and to minimize both over- and under-segmentation, a laminin-based segmentation filter was applied. Filter values of 5, 10, or 20 were used consistently across samples. One IIA soleus control sample required a filter value of 10 to improve segmentation accuracy, whereas the corresponding denervated left muscle sample required a filter value of 20. Pixel size was uniformly set to one. To ensure accurate fiber detection, Minimum and maximum

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fiber area thresholds were then manually optimized for each muscle fiber-type and animal by inspecting images to confirm that all true fibers were correctly identified; objects that were too small or too large to represent genuine fibers were excluded. Minimum thresholds ranged from 100–3,000μm² and maximum thresholds from 5,000–20,000μm², and these thresholds were applied identically to both the denervated and contralateral control sides. These parameters were not blinded, as knowledge of the condition was necessary to maintain consistency in segmentation across matched samples.

Additional segmentation parameters included a fixed maximum eccentricity of 0.99 and a minimum convexity of 0.7. Because the SMASH algorithm uses normalized fluorescence intensity values, fiber-type threshold is unitless; for this study, it was empirically adjusted within a range of 5–20. Threshold selection was guided by visual inspection of the fluorescence images to ensure all genuine fibers were highlighted, while artifact and background were excluded. Once an optimal threshold was identified for a given fiber type, it was held constant across paired conditions whenever possible. Paired left-right fiber-type threshold consistency was documented for each animal, muscle/site group, and fiber type. Of 34 paired comparisons, 30 used identical fiber-type thresholds. Four paired comparisons differed by 5 threshold units. Among these four nonzero adjustments, the percent difference relative to the contralateral/right threshold ranged from 50–100%, calculated as |left threshold − right threshold| / right threshold × 100. This percentage appears large because thresholds were small discrete values. In absolute terms, each nonzero adjustment represented the same 5-unit threshold

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difference. Percent threshold differences were used only as descriptive threshold-consistency metrics and not for statistical analysis.

Minimum Feret diameter (mFD) was calculated for each segmented fiber and matched to positively identified fiber-types. Minimum Feret diameter is a shape-based morphometric measure defined as the smallest possible distance between two parallel lines tangent to the fiber boundary when the object is rotated through all possible angles. Unlike simple width measurements, mFD is independent of fiber orientation and less affected by oblique sectioning, making it a robust metric to estimate muscle fiber size in cross-sectional images. For all fiber-types except IIX, classification was based on positive immunolabeling. IIX fibers were classified by exclusion, defined as segmented unlabeled fibers.

RNA Sequencing. To collect lateral and MTA muscles from denervated and contralateral internal-reference vocal folds, the mucosa was removed following methods previously described by our laboratory.(35) Once the TA was exposed, the boundary between the medial and lateral TA was visually identified under magnification. To isolate the MTA, the LTA was detached from the arytenoid cartilage. The arytenoid cartilage and MTA were then peeled as a unit from the LTA. Anteriorly, the MTA was separated from the arytenoid cartilage and placed in cold PBS.

Fine-tipped forceps and microscissors were used to dissect connective tissue attachments between the PCA and surrounding structures (cricoid cartilage and other intrinsic laryngeal muscles) and placed in cold PBS. The gastrocnemius and soleus muscles were transected at their midpoints. The proximal portions were embedded in OCT for complementary immunolabeling. The distal portions were cleared of fascia and

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nerves and placed in cold PBS. The MTA, LTA, PCA, distal gastrocnemius, and distal soleus muscle samples (all in cold PBS) were placed separately in individual 30mm cell culture dishes. Each sample was processed individually; samples were not pooled across animals, sides, or muscle types.

The tissues were minced, triturated using a 1000µL pipette, transferred to conical tubes, and further triturated in Buffer RLT containing β-mercaptoethanol (QIAGEN, Hilden, Germany). The samples were stored at -80°C. Total RNA was isolated from denervated and control muscle tissues using the RNeasy Mini Kit (QIAGEN, Hilden, Germany) and RNA integrity was verified using an Agilent Bioanalyzer. Samples ≤1ng were reverse transcribed into complementary DNA (cDNA) using the SMART-Seq HT Kit (Takara Bio USA, Mountain View, CA, USA), with 13 PCR cycles. cDNA was quantified using the Invitrogen Quant-iT system (Invitrogen/Thermo Fisher Scientific, Waltham, MA, USA) and diluted to 0.3ng/µL for library preparation.

Library Preparation and Sequencing. Libraries were prepared with Nextera XT (Illumina, San Diego, CA, USA), indexed, and quality-checked on an Agilent TapeStation. Libraries were pooled equimolarly and sequenced on an Illumina NovaSeq X+ platform (Illumina, San Diego, CA, USA; paired-end, 100-cycle configuration).

RNA-Seq Processing and Differential Expression**.** Raw reads were trimmed (Cutadapt),(36) quality-checked (FastQC),(37) aligned to the Rattus norvegicus genome (BN7.2) using STAR,(38), and quantified with HTSeq.(39) Differential expression was computed in Rosalind® (OnRamp Bio, San Diego, CA, USA) using DESeq2(40) with Relative Log Expression normalization and covariate correction by sex. Significance was defined as false discovery rate (FDR) - adjusted $p \leq 0.05$). RNA and library quality

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were high across samples: RIN ≥ 8 (most ≥ 9), Q30 ≈ 98%, 94–97% alignment, detection of ~16,000–19,700 genes per sample, and ~69–80% reads retained in count matrices. Sample-level RNA integrity and sequencing quality metrics are provided in **Supplementary Table 2**. Given our modest sample size (n=3 per condition), we emphasized interpretation at the pathway level, consistent with the gene set enrichment framework, which detects coordinated changes across functionally related genes.(41)

Pathway Enrichment. Pathway enrichment analyses were conducted using the Rosalind platform (OnRamp Bio). Gene lists were derived from RNA-seq data based on fold-change thresholds of ±4 for primary analysis and ±1.5 for exploratory analysis; statistical significance was determined by FDR - adjusted p≤0.05. Enrichment was evaluated using the BioPlanet 2019 database(42) which integrates curated biological pathways across diverse functional domains. Pathways were considered significantly enriched if they met the FDR significance threshold within each tissue-specific gene set.

As a complementary pathway analysis, KEGG enrichment was performed using Enrichr on tissue-specific ±4-fold-change DEG lists. Upregulated and downregulated gene lists were analyzed separately for each muscle. Enrichr-derived p-values and q-values were exported and retained as the primary enrichment statistics. Full exported KEGG/Enrichr results are provided in **Supplementary Table 3.**

Statistical Analyses. All mFD data were analyzed using GraphPad Prism version 10 (GraphPad Software, Boston, MA, USA). Percent (%) change in mFD was calculated by comparing reduction in fiber size on the denervated side relative to control. Normality was confirmed for all groups using the Shapiro–Wilk test ($p>0.05$), and homogeneity of variance was verified using the Brown–Forsythe test; no significant differences in

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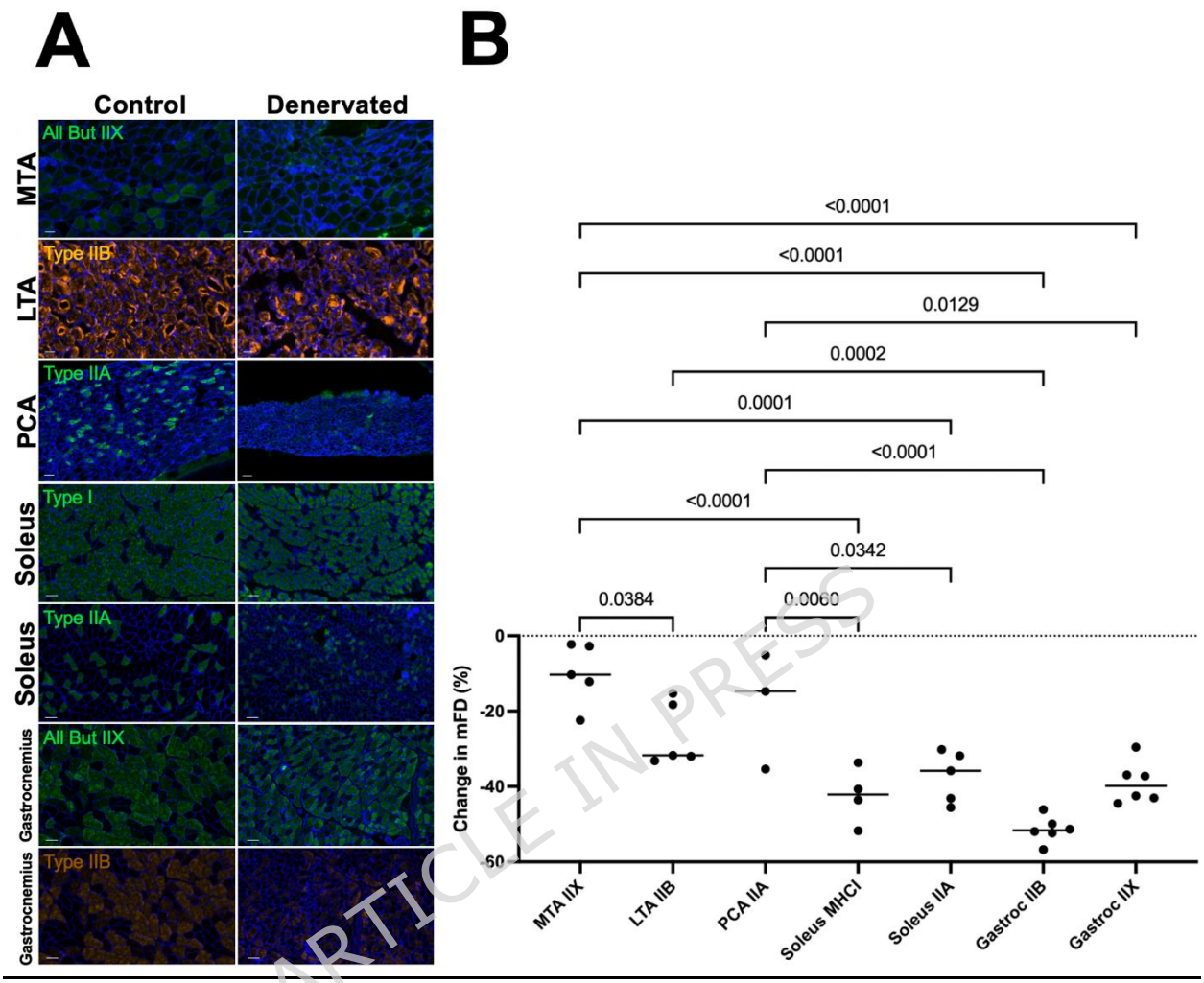
standard deviation were observed across muscle fiber-types ($p>0.05$). A one-way ANOVA was performed to assess differences across muscle fiber-types. Post-hoc comparisons were conducted using Tukey's multiple comparisons test, with significance defined as adjusted $p \leq 0.05$ (**Supplementary Table 4**).

Results

Minimum Feret diameter was preserved in laryngeal muscles compared to matched hindlimb muscles. Denervation-induced atrophy varied across fiber-type and anatomical region. Less atrophy was observed in laryngeal muscles than hindlimb muscles, even among fibers of the same type. Specifically, MTA putative IIX fibers atrophied significantly less than gastrocnemius IIX fibers (adjusted $p<0.0001$), PCA IIA fibers less than Soleus IIA fibers (adjusted $p=0.034$), and LTA IIB fibers less than gastrocnemius IIB fibers (adjusted $p<0.001$). Notably, numerically greater LTA IIB atrophy was observed in females (n=3) than males (n=2) (−32.3% vs. −16.8%), a trend not observed in any other muscle or fiber-type. To confirm this sex imbalance did not drive the overall effect, we repeated the analysis for LTA IIB after excluding the two males: the relationship remained unchanged - LTA IIB was still significantly less atrophic than gastrocnemius IIB fibers (adjusted $p=0.015$) and more atrophic than MTA putative IIX/IIX-like fibers (adjusted $p=0.005$). These results suggest relative resistance of

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laryngeal muscles to denervation-induced atrophy (**Figure 1A** and **1B).**



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Within the larynx, LTA IIB fibers in the larynx atrophied significantly more than MTA putative IIX/IIX-like fibers (adjusted p=0.038), highlighting fiber subtype–specificity within the same organ. In the hindlimb, gastrocnemius IIB fibers atrophied more than all other fiber-types. However, these differences did not achieve statistical significance, indicating variability in susceptibility may not be uniform across all fiber subtypes.

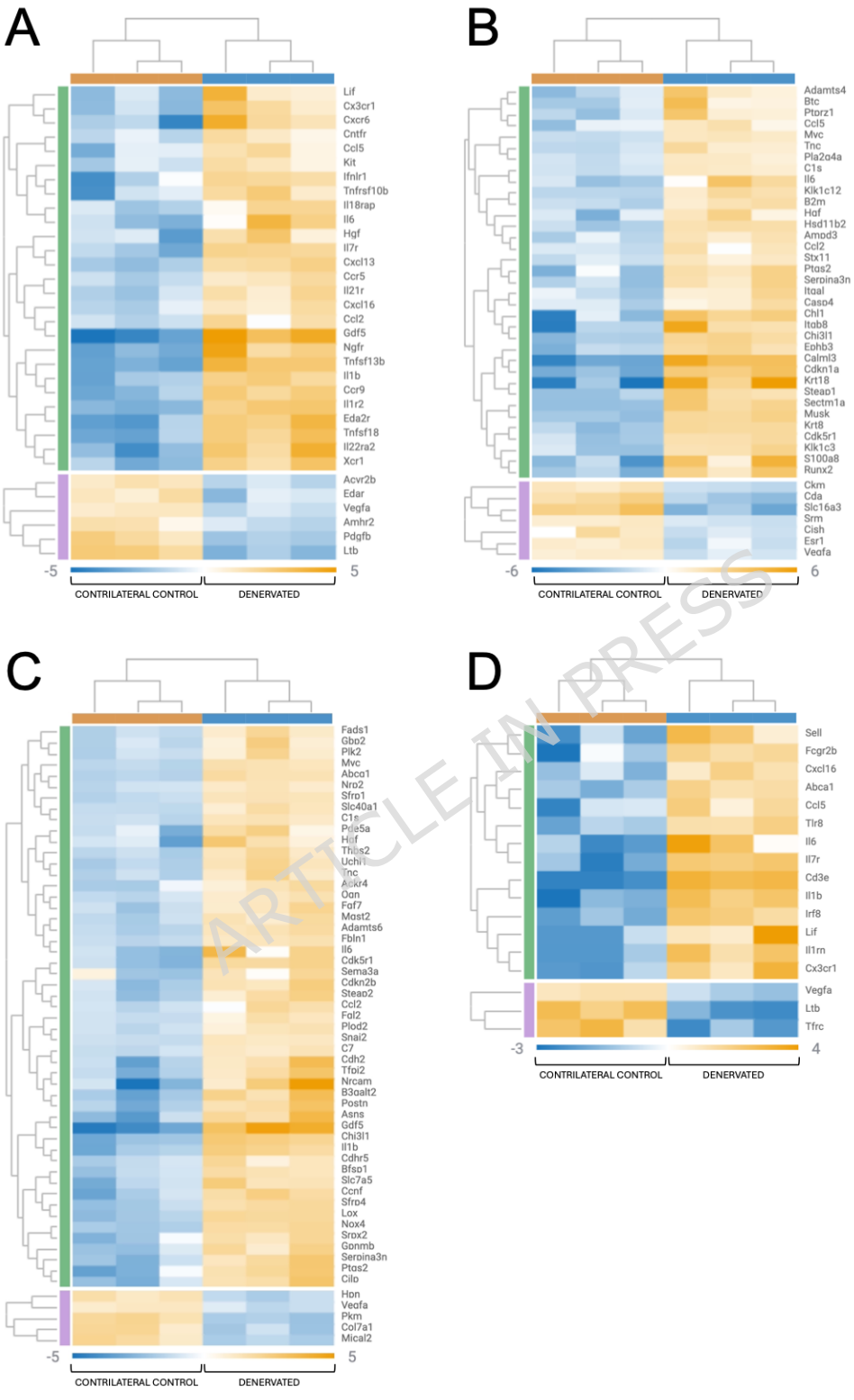
Transcriptomic Pathway Enrichment Highlighted Region-Specific Denervation Responses Across Muscle Types. To investigate molecular mechanisms underlying region-specific denervation, we performed bulk RNA sequencing. To provide an

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overview of differential expression patterns across all muscles, volcano plots were generated for each tissue comparison and are provided in **Supplementary Figure 1**. These plots show the distribution of log2 fold-change and adjusted p-values across detected genes and illustrate the broader transcriptional response in hindlimb muscles compared with the more restricted response observed in laryngeal muscles. This analysis revealed significant pathway enrichment at ±4 fold-change threshold in the gastrocnemius and PCA muscles. In gastrocnemius, enriched pathways included Oncostatin M signaling (FDR=1.2×10^{-4}; **Figure 2A**)**,** cytokine–cytokine receptor interaction (FDR=5.38×10^{-3}; **Figure 2B**), TGF-β regulation of extracellular matrix (FDR=3.66×10^{-2}; **Figure 2C**), and leptin influence on immune response (FDR=4.11×10^{-2}; **Figure 2D**).

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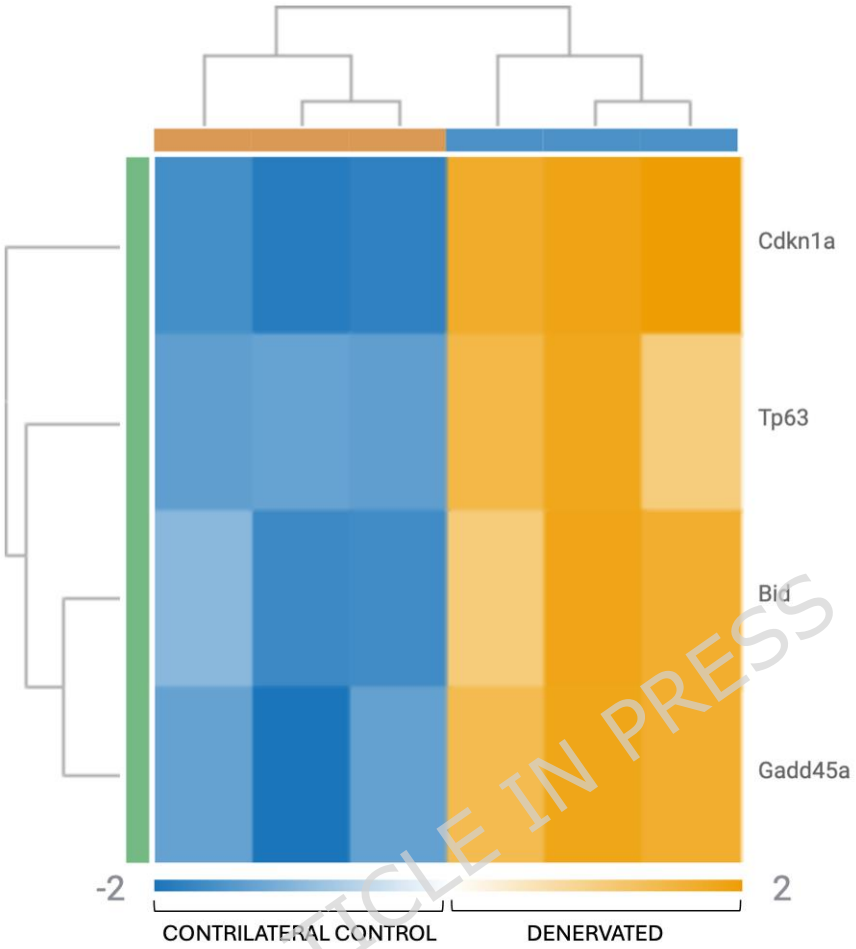
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In PCA, the TP53 network was significantly enriched (FDR=1.0×10^{-3}; **Figure 3**). The PCA TP53-associated signal was driven by a small set of upregulated genes, including Cdkn1a, Tp63, Bid, and Gadd45a. In complementary KEGG/Enrichr analysis, the PCA upregulated ±4-fold DEG list was also significantly enriched for the p53 signaling pathway, supporting a PCA-specific TP53-associated stress-response signature.

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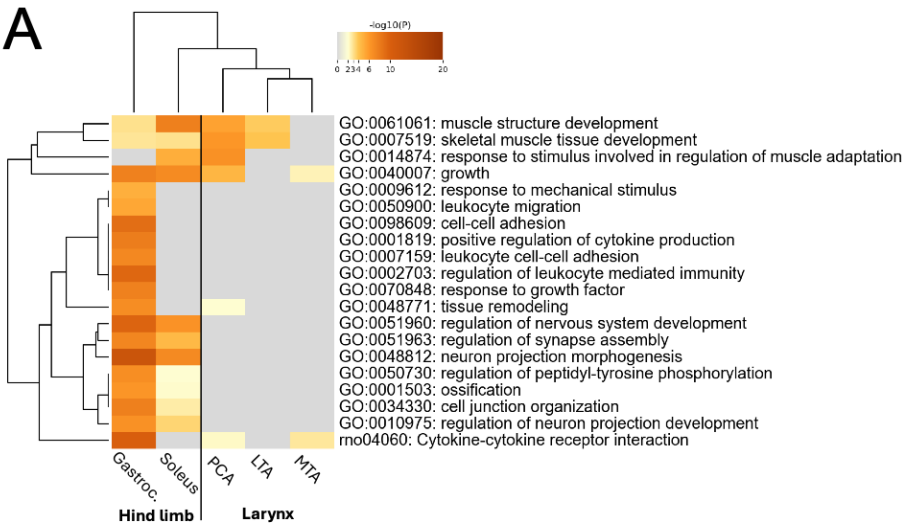
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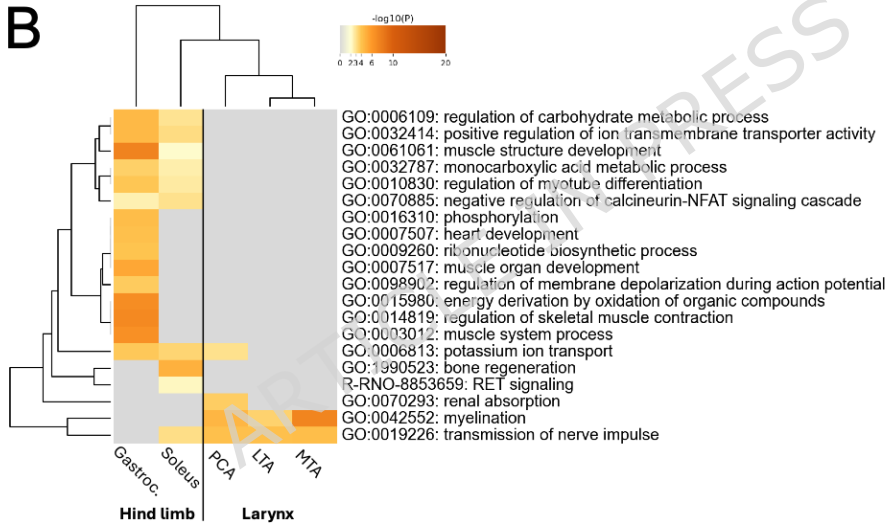
Within the primary BioPlanet analysis, no significant pathways were identified in soleus, MTA, or LTA tissues at the ±4-fold threshold. However, the underlying up- and downregulated gene lists at this threshold are provided by tissue and direction of regulation in **Supplementary Table 5.**

To directly compare organ-level transcriptional responses, we next examined Gene Ontology enrichment across the two hindlimb muscles and the three intrinsic laryngeal muscles. This analysis allowed shared and tissue-specific biological processes to be visualized across all studied muscles rather than within individual pathway heatmaps alone. The ±4 fold change gene sets were further analyzed in Metascape,(43) which generated GO-term heatmaps (**Figure 4A and B**).

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GO analysis revealed both conserved and organ-selective responses. Upregulated gene lists from both hindlimb and laryngeal muscles showed enrichment for muscle structure development, skeletal muscle development, and stimulus-response pathways related to muscle adaptation. In contrast, hindlimb muscles showed broader enrichment of pathways related to nerve impulse transmission, inflammatory signaling, extracellular matrix remodeling, and metabolic adaptation, whereas laryngeal muscles showed more restricted and muscle-selective enrichment.

Complementary KEGG/Enrichr enrichment at the ±4-fold-change threshold supported pathway-level RNA-seq interpretation. Gastrocnemius showed the broadest KEGG/Enrichr enrichment, particularly among upregulated genes, with significant enrichment of cytokine-cytokine receptor interaction, cell adhesion molecule interaction, IGSF CAM signaling, and chemokine signaling pathways. In contrast, laryngeal muscles showed more selective enrichment. MTA upregulated genes showed significant cytokine-cytokine receptor interaction enrichment, whereas PCA upregulated genes showed significant p53 signaling pathway enrichment. A compact visualization of FDR-significant and exploratory KEGG/Enrichr terms is provided in **Supplementary Figure 2**.

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Gene-level signatures of developmental and craniofacial myosins at the ±4-fold threshold. To complement pathway-level analyses, we queried canonical developmental and craniofacial MyHC transcripts. In gastrocnemius, *Myh15* (≈13-fold) and *Myh3* (≈8-fold) were markedly upregulated with very low adjusted p-values. In soleus, *Myh8* (≈13-fold) was strongly induced. Within the larynx, LTA showed the largest single increase overall with *Myh15* (≈14.5-fold) and in PCA, both *Myh15* (≈8-fold) and *Myh8*

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(≈5-fold) were significantly elevated (all FDR ≤ 0.05, |FC| ≥ 4). These gene-level patterns indicate a robust re-expression of developmental/perinatal myosins (e.g., *Myh3*, *Myh8*) and induction of craniofacial-associated myosins (e.g., *Myh15*) after denervation, with muscle-specific magnitudes that mirror the organ-context differences observed at the pathway level.

At an exploratory fold-change threshold of ±1.5, a broader set of pathways emerged across tissues. Extensive enrichment was observed in the gastrocnemius for mitochondrial and metabolic processes, including the tricarboxylic acid (TCA) cycle and respiratory electron transport (FDR=3.3×10^{-7}), electron transport chain (FDR=6.3×10^{-6}), oxidative phosphorylation (FDR=8.8×10^{-5}), and pyruvate metabolism (FDR=2.25×10^{-2}). Additional enriched terms included phospholipid metabolism (FDR=2.78×10^{-3}), insulin signaling (FDR=4.4×10^{-3}), AMPK signaling (FDR=1.56×10^{-2}), mTOR signaling (FDR=2.62×10^{-2}), and muscle contraction (FDR=2.25×10^{-2}). A similar oxidative profile was observed in the soleus, with strong enrichment for the TCA cycle and respiratory electron transport (FDR=8.0×10^{-9}), pyruvate metabolism (FDR=9.7×10^{-4}), mitochondrial fatty acid β-oxidation (FDR=6.5×10^{-5}), and muscle contraction (FDR=7.99×10^{-3}). In contrast, more modest changes were observed in laryngeal muscles. Selective enrichment was observed in the PCA for electron transport chain, VEGF signaling, TSH-regulated gene expression, and ATP biosynthesis (all FDR=7.75×10^{-3}), as well as the TCA cycle (FDR=1.07×10^{-2}). Modest enrichment was observed in the MTA for the TCA cycle (FDR=1.79×10^{-2}) and muscle contraction (FDR=4.46×10^{-2}). No significant pathways were observed in the

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LTA at this threshold. A complete list of enriched pathways at the ±1.5 fold-change threshold is provided in **Supplementary Table 6.**

Discussion

This preliminary study provides translational insight into laryngeal atrophy biology by showing that denervation outcomes were governed primarily by organ context with fiber subtype as a secondary modifier. Across matched fast subtypes, laryngeal muscles underwent less atrophy with muted transcriptomic remodeling compared to hindlimb muscles. In parallel, these data refine the experimental value of the rat laryngeal model: not as a fiber-type analog of the human TA, but as a mechanistic analog that captures organ-specific protective/remodeling programs relevant to human disease. Denervation resulted in divergent structural and transcriptional responses across muscles that differ by fiber subtype and anatomical context. Using fiber size quantified by mFD, laryngeal muscles were relatively resistant to atrophy compared to hindlimb muscles, even when fiber classes were matched. Less atrophy was observed in the MTA (primarily type IIX) than gastrocnemius IIX fibers, LTA (primarily type IIB) fibers atrophied less than gastrocnemius IIB fibers, and PCA (enriched for type IIA/IIX) fibers were less atrophic than soleus IIA fibers. Within the larynx, vulnerability still varied by subtype - greater atrophy was observed in the LTA (enriched for type IIB) than MTA IIX - indicating fiber subtype contributes to, but does not dictate, outcomes within the same organ. Prior work demonstrated hybrid MyHC phenotypes in rat TA subdivisions and identified MYH13/EO as a specialized myosin expressed in laryngeal and other craniofacial muscles, consistent with a myosin repertoire uncommon in limb

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musculature.(17,28) We did not quantify MYH13/EO in the present cohort, but the literature-supported presence of EO-associated isoforms in LTA(17) and their absence in hindlimb emphasize that organ context includes a distinct myosin landscape which may contribute to the attenuated, selective transcriptional remodeling we observed in laryngeal muscles. Taken together, these observations support organ context as a major determinant of denervation outcome while acknowledging a measurable contribution of fiber subtype to relative susceptibility.

Transcriptomic analyses paralleled structural findings and highlighted how detectable signal scales with analytical stringency. At the stringent ±4-fold threshold, enrichment of inflammatory and extracellular matrix–associated pathways was observed in the gastrocnemius, including Oncostatin M, cytokine-cytokine receptor interaction, and TGF-β regulation of the extracellular matrix-processes known to regulate regeneration and extracellular matrix deposition.(44–48) TP53 network enrichment was observed in the PCA, consistent with prior reports linking TP53 activation to skeletal muscle atrophy, mitochondrial stress, and fibrotic remodeling.(49–51) No pathways reached significance in soleus, MTA, or LTA at this threshold. At the exploratory ±1.5 threshold, extensive enrichment for oxidative metabolism and mitochondrial pathways were observed in the gastrocnemius and soleus including the tricarboxylic acid cycle, electron transport chain, oxidative phosphorylation, pyruvate metabolism, and fatty acid β-oxidation, along with insulin and AMPK signaling - key regulators of energy homeostasis in skeletal muscle.(52) Laryngeal responses were attenuated and muscle-selective: enrichment for electron transport chain, ATP biosynthesis, VEGF signaling, and TSH-regulated gene expression were observed in the PCA. Modest mitochondrial

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and contraction terms were observed in the MTA. The LTA remained largely unchanged. Notably, in the PCA, oxidative and ATP biosynthesis pathways were predominantly downregulated even as TP53-related genes were upregulated at the ±4 fold change threshold, possibly indicating process-specific and threshold-dependent remodeling rather than a single uniform program. Downregulation of oxidative phosphorylation and TCA cycle activity aligns with prior evidence of impaired energy metabolism during muscle disuse and atrophy.(53) VEGF signaling was also reduced in PCA, consistent with diminished angiogenic signaling in other denervated hindlimb models.(53,54) A mixed pattern of TSH regulated gene expression was observed, consistent with reports of thyroid hormone receptor expression in muscle and its minor role in differentiation markers such as myogenin.(55)

Metascape distinguished conserved from muscle-specific components of the denervation response. Gene Ontology heatmaps revealed shared functional themes among upregulated genes, particularly pathways related to muscle structure development and stimulus-response processes. Downregulated genes included conserved suppression of nerve impulse–related pathways across multiple muscles, whereas BioPlanet analysis indicated gastrocnemius uniquely engaged broader inflammatory, extracellular matrix, and metabolic programs. These patterns suggest, although certain adaptive signatures transcend fiber class, the magnitude and breadth of remodeling remain strongly tissue-specific. Complementary KEGG/Enrichr enrichment at the ±4-fold-change threshold supported pathway-level RNA-seq interpretation. Gastrocnemius showed the broadest KEGG/Enrichr enrichment, particularly among upregulated genes, with significant enrichment of cytokine-cytokine

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receptor interaction, cell adhesion molecule interaction, IGSF CAM signaling, and chemokine signaling pathways. In contrast, laryngeal muscles showed more selective enrichment. MTA upregulated genes showed significant cytokine-cytokine receptor interaction enrichment, whereas PCA upregulated genes showed significant p53 signaling pathway enrichment.

Several features of the laryngeal microenvironment may explain resistance to atrophy relative to hindlimb muscles - these mechanisms were not investigated and remain hypothetical. In limb muscles such as the soleus, acetylcholine receptor (AChR) reactivity declines rapidly - falling to ~54% and ~35% of baseline by 33 and 57 days post-denervation.(56) In intrinsic laryngeal muscles such as the thyroarytenoid (TA), AChR organization remains largely intact for approximately four weeks and still retains ~70% reactivity at 10 weeks.(57) Together, these findings suggest prolonged integrity of neuromuscular junction (NMJ) components in intrinsic laryngeal muscles relative to limb muscles after denervation, which may delay initiation of atrophic signaling.

Compensatory activation patterns unique to the larynx such as cricothyroid engagement during attempted phonation may help preserve mechanical loading on denervated fibers. Ipsilateral Superior Laryngeal Nerve-mediated activation of the cricothyroid was observed in a canine RLN paralysis model, which elongates and tenses the vocal fold despite TA denervation.(58) This maintained tension could provide an anti-atrophic mechanical signal not present in simpler systems like the hindlimb, potentially contributing to reduced atrophy observed in intrinsic laryngeal muscles. In addition, satellite cell–mediated myonuclear addition has been documented in the TA following RLN injury, including a contralateral response consistent with altered

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workload. Approximately ~6% of denervated TA fibers contained newly incorporated BrdU-positive nuclei at one week, compared to ~0.1% in controls. In addition, ~3–4% BrdU-positive fibers were observed in the contralateral TA, suggesting compensatory loading.(59) These findings indicate both local remodeling and cross-laryngeal compensation may contribute to preserved TA fiber size and structural integrity. Collectively, these data provide putative mechanisms underlying decreased structural and transcriptomic remodeling observed in laryngeal muscles.

Larynx-specific neural anatomy may also contribute to the attenuated response observed after RLN transection. Although the RLN provides primary motor innervation to most intrinsic laryngeal muscles, intralaryngeal communicating branches and SLN–RLN anastomoses have been described. These laryngeal nerve communications can alter innervation patterns of the intrinsic laryngeal muscles and may affect interpretation of paralysis or denervation models.(60) Recent *ex vivo* animal work further suggests the cricothyroid and internal thyroarytenoid/vocalis muscles can show bidirectional coactivation in the absence of external central neural control, supporting the possibility of a localized intrinsic laryngeal neuronal network.(61) The present RLN transection model should be interpreted as a model of RLN injury rather than proof of complete elimination of all possible laryngeal neural influence. These neural particularities may contribute to residual trophic signaling, altered local activity, or delayed atrophic remodeling and should be considered organ-specific features of the larynx rather than purely technical confounds. Future studies directly comparing RLN-only injury with combined RLN/SLN interruption, where feasible, are required to determine the contribution of these neural pathways to laryngeal muscle preservation after

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denervation. Specialized neuromuscular organization within the intrinsic laryngeal muscles may further complicate interpretation of fiber-subtype responses. Recent evidence emphasizes intrinsic laryngeal muscles differ from limb muscles in MyHC composition and neural organization, including specialized/mixed MyHC phenotypes, dense neural representation, multiple neural terminals, and intramuscular neural elements.(62) More recent human anatomical work in the vocalis/internal thyroarytenoid region described abundant nerve fibers, widespread sensory and motor synaptic structures, possible intramuscular neurons, and multi-synaptic muscle fibers within the pars interna of the thyroarytenoid muscle.(63) Although these findings were reported in human tissue and were not directly assessed in the present rat model, they highlight a larynx-specific neuromuscular architecture that may differ from limb muscle and could influence how intrinsic laryngeal muscles respond to RLN injury. If comparable local neuromuscular features are present or active in the rat larynx, they could contribute to residual trophic signaling, altered local activity, or delayed atrophic remodeling. This possibility may be relevant to the relative preservation of MTA putative IIX/IIX-like fibers observed here and to the greater resistance of MTA compared with the more atrophic LTA, but remains speculative and should be tested directly in future studies. This interpretation is also consistent with prior human PCA data showing RLN injury is associated with time-dependent MyHC remodeling, including reduced MHC-I-positive fibers, increased MHC-II-positive fibers, increased MHC-IIx and developmental MyHC transcripts, hybrid fibers, and evidence suggesting residual or subclinical innervation after denervation. (64)

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Within the larynx, we also found type IIB fibers atrophied more than putative IIX or IIA fibers. These findings align with prior work suggesting disproportionate denervation vulnerability of fast-glycolytic type IIB fibers. For example, greater reduction in fiber size was observed in IIB fibers compared to IIA or IIX following denervation in the gastrocnemius.(65) Importantly, the human thyroarytenoid lacks type IIB fibers and contains both type I and type II fibers;(66) therefore, the rat larynx is not a fiber-type analog of the human larynx. Nevertheless, our data support the rat larynx as a mechanistic analog to study organ-specific protective and remodeling responses to denervation. In this framework, the fiber-type paradigm is refined rather than overturned - fiber subtype influences vulnerability, but does not uniformly determine outcomes across organs.

Regarding transcriptomic analyses, the dual-threshold approach was informative. The ±4 analysis provided a conservative, high-specificity view that cleanly distinguished robust hindlimb remodeling from largely quiescent laryngeal profiles. The exploratory ±1.5 analysis increased pathway recovery in expected directions - extensive oxidative metabolism in hindlimb; selective oxidative and signaling terms in PCA; modest mitochondrial terms in MTA and the LTA was largely unchanged. The concordance between fiber size changes and pathway behavior across thresholds suggests minimal transcriptomic changes observed in laryngeal muscles were biological in origin rather than technical artifact. Interpreted alongside morphometry, these data support the concept that transcriptional programs associated with developmental/craniofacial myosins can be engaged without obligate convergence to a uniform atrophy phenotype. Hindlimb muscles displayed broad pathway activation and strong developmental MyHC

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induction, whereas laryngeal muscles exhibited selective (PCA, LTA) rather than global induction, consistent with their muted pathway remodeling and relative structural preservation.

This study has several important limitations that constrain interpretation and reinforce its preliminary, hypothesis-generating nature. First, the cohort size was modest, limiting statistical power and precluding robust sex-specific analyses. Although sex was included as a covariate in the RNA-seq analysis, the study was not powered to determine whether denervation responses differed between males and females. Sex distribution was also uneven across some morphometric comparisons; for example, greater atrophy was observed in female LTA IIB fibers than males, although this trend was not consistent across other muscles and LTA IIB fibers remained significantly less atrophic than gastrocnemius IIB and more atrophic than MTA putative IIX fibers after excluding males. These observations should therefore be considered exploratory. Future studies should include larger, sex-balanced cohorts designed a priori to test sex as a biological variable. Second, the study used the contralateral side as a within-animal reference. This design reduces inter-animal variability and is useful for preliminary paired comparisons, but it does not replace an independent uninjured control group. Unilateral laryngeal denervation may alter contralateral muscle loading, compensatory recruitment, or gene expression, and intrinsic left–right differences in laryngeal muscle gene expression have been reported in uninjured rats, especially in adductor muscles and adult males.(67)

Some portion of the observed differences could reflect contralateral compensation or baseline laterality rather than denervation alone. Future studies should

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include independent uninjured controls, counterbalanced injury side, and/or bilateral injury paradigms where appropriate. A further limitation is that RLN transection cannot exclude all possible laryngeal neural influence, including SLN–RLN anastomoses, intralaryngeal communicating branches, or local intralaryngeal neuromuscular networks; these features were not directly tested here and may contribute to residual trophic signaling, altered local activity, or delayed atrophic remodeling.

Finally, RNA-seq was performed with three biological replicates per condition. As such, these data should be considered preliminary and exploratory with limited power to detect small effects and no ability to assess temporal dynamics. To mitigate these constraints, we ensured high RNA and library quality (RIN ≥ 8; Q30 ≈ 98%; 94–97% alignment; ~16,000–19,700 genes detected; ~69–80% of reads retained), applied DESeq2 with Relative Log Expression normalization and FDR control, and emphasized pathway-level interpretation (BioPlanet 2019 and Metascape) rather than isolated gene-level changes. This approach is recommended for modestly replicated bulk RNA-seq datasets.(41) It is well-established underpowered bulk RNA-seq studies ($N \leq 4$) are prone to inflated effect sizes and reduced gene-level reproducibility - increased fold-change thresholds alone do not compensate for limited replication. As such, we applied a stringent ±4× threshold to enrich for robust signals and prioritized pathway-level analyses to capture biologically coherent trends appropriate for this sample size.(68) Because bulk profiling captures tissue-level signals, the RNA-seq reflects mixtures of myofibers, fibroadipogenic cells, vasculature, and immune populations; cell-type–specific and spatially restricted responses cannot be resolved here, and single-nucleus

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or spatial transcriptomics will better attribute signals to individual fiber classes, NMJ regions, and interstitial compartments.

A further limitation is that putative IIX/IIX-like fibers were identified operationally by absence of BF-35 immunoreactivity rather than by direct detection of an IIX-specific epitope. BF-35 is an “all-but-IIX” antibody that labels MyHC isoforms other than IIX and is commonly used to identify putative IIX/IIX-like fibers by negative staining. This approach is commonly used in rodent MyHC immunolabeling panels, but is imperfect in intrinsic laryngeal muscles where hybrid MyHC expression and EO/MYH13-containing fibers may occur. Therefore, the IIX category should be interpreted as putative IIX/IIX-like rather than definitive pure IIX. Future studies incorporating direct MYH13/EO labeling, additional MyHC markers, and single-fiber validation will be needed to resolve hybrid-fiber composition more precisely.

Some hybrids were therefore likely classified by their dominant detectable epitope or subsumed within the operational putative IIX/IIX-like category. However, the primary cross-organ atrophy pattern (laryngeal < hindlimb) and attenuated laryngeal transcriptomic remodeling were also observed in non-LTA fiber-matched contrasts, including MTA putative IIX versus gastrocnemius putative IIX and PCA IIA versus soleus IIA. The main conclusion does not depend solely on LTA IIB classification in this preliminary dataset.

Sections were sampled from defined anatomical regions (e.g., PCA at cricoid insertion; lateral head of gastrocnemius near the surface) and SMASH parameters were applied identically to paired sides but were not blinded; these choices could modestly influence absolute morphometrics. Using the contralateral side as the reference can

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also introduce compensatory loading on the uninjured side, potentially underestimating absolute denervation effects. For enrichment analysis, gene lists were interrogated with BioPlanet (human-curated) via ortholog mapping and with Metascape GO using rat annotations; pathway calls are therefore treated as directional and supportive rather than definitive, and concordant results across BioPlanet and GO terms increase confidence in broad themes rather than specific edges. Finally, we did not directly quantify NMJ integrity, in vivo loading, or satellite-cell activity in these cohorts—mechanistic inferences about these processes remain untested—and species and organ differences remain relevant; rat intrinsic laryngeal muscles differ from human counterparts, and observations at this single post-injury time point may not generalize across developmental stages or injury windows. In addition, intrinsic laryngeal motor units are exceptionally small (e.g., in rat, PCA ≈4–5 fibers per motoneuron and CT ≈8 fibers per motoneuron)(69) which complicates inference of motor-unit survival or remodeling from single cross-section myosin labeling alone. Resolving motor-unit-level rearrangements requires retrograde motoneuron tracing combined with AChE-positive endplate mapping and/or longitudinal single-fiber MyHC profiling to capture territory-level changes after denervation. This study sought to provide organ-context and testable hypotheses for follow-up studies with expanded cohorts, time courses, and cell-resolved approaches.

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Conclusion

In this preliminary study, organ context - not fiber-type alone - governed denervation outcomes. Laryngeal muscles resisted atrophy and exhibited attenuated,

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selective transcriptional remodeling relative to hindlimb muscles. Subtype differences within the larynx revealed persistent vulnerability of fast-glycolytic type IIB fibers. Because human TA muscle lacks type IIB fibers, the rat larynx should not be treated as a fiber-type analog. However, it remains a mechanistic analog to query conserved, organ-specific protective and remodeling responses. This perspective is foundational for subsequent hypothesis-driven studies of larynx-specific mechanisms such as NMJ persistence, loading-dependent signaling, and stress-responsive programs and offers a path toward targeted interventions to preserve laryngeal function after denervation.

Abbreviations

AChR — Acetylcholine receptor

AMPK — AMP-activated protein kinase

ANOVA — Analysis of variance

BF-35 — “All but IIX” myosin heavy chain antibody

BF-F3 — Myosin heavy chain IIB antibody

cDNA — Complementary DNA

DEG — Differentially expressed gene

DESeq2 — Differential expression package (R/Bioconductor)

ECM — Extracellular matrix

EO — Extraocular

FDR — False discovery rate

GO — Gene Ontology

HTSeq — High-Throughput Sequencing framework

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IACUC — Institutional Animal Care and Use Committee

IIA/IIB/IIX — Myosin heavy chain fiber subtypes

LTA — Lateral thyroarytenoid

MyHC — Myosin heavy chain

MTA — Medial thyroarytenoid

mFD — Minimum Feret diameter

mTOR — Mechanistic target of rapamycin

NMJ — Neuromuscular junction

OCT — Optimal cutting temperature compound

PCA — Posterior cricoarytenoid

PBS — Phosphate-buffered saline

RLN — Recurrent laryngeal nerve

RNA-seq — RNA sequencing

SRA — Sequence Read Archive

STAR — Spliced Transcripts Alignment to a Reference

TA — Thyroarytenoid

TCA — Tricarboxylic acid cycle

TP53 — Tumor protein 53

VEGF — Vascular endothelial growth factor

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Declarations

All animal procedures were approved by the NYU Grossman School of Medicine IACUC (protocol #202400004) and conducted in accordance with institutional and national

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guidelines. Reporting of animal procedures followed ARRIVE guidelines, where applicable.

Consent for publication

Not applicable.

Availability of data and materials

All processed RNA-seq outputs and morphometry datasets generated in this study are deposited on Figshare (https://doi.org/10.6084/m9.figshare.30880157). Raw FASTQ files are deposited in GEO under accession number GSE325272.

Competing interests

The authors declare no competing interests.

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Authors' contributions

Conceptualization: Gary G. Gartling (GG); Ryan C. Branski (RCB).

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Methodology: GG; Masayoshi Yoshimatsu (MY); Aaron M. Johnson (AMJ); Michael Garabedian (MG).

Investigation: GG (surgeries; imaging; morphometry); MY (surgeries; RNA-seq tissue collection); Renjie Bing (RB) (PCR; surgery assistance); Elizabeth Hary (EH) (larynx sectioning, writing); Vivian Didomenico (VD) (tissue sectioning).

Resources: AMJ (immunolabeling protocol guidance; SMASH usage support).

Data curation: GG.

Formal analysis: GG (morphometry, RNA-seq); MG (RNA-seq analysis and interpretation).

Software/Code: GG (QuPath/SMASH parameters; DESeq2 scripts).

Visualization: GG.

Writing—original draft: GG.

Writing—review & editing: MY; EH; RB; VD; MG; AMJ; RCB.

Supervision: RCB; AMJ.

Project administration: GG; RCB.

All authors meet journal authorship criteria, approved the final manuscript, and are accountable for the work.

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References

1. Clavel S, Coldefy AS, Kurkdjian E, Salles J, Margaritis I, Derijard B. Atrophy-related ubiquitin ligases, atrogin-1 and MuRF1 are up-regulated in aged rat Tibialis Anterior muscle. Mech Ageing Dev. 2006;127(10):794–801. doi:10.1016/j.mad.2006.07.005

2. Kostrominova TY. Skeletal Muscle Denervation: Past, Present and Future. Int J Mol Sci. 2022 Jan;23(14):14. doi:10.3390/ijms23147489

3. Ready A, Barnes A. Complications of thyroidectomy. J Br Surg. 1994;81(11):1555–6.

4. Dewan K, Cephus C, Owczarzak V, Ocampo E. Incidence and implication of vocal fold paresis following neonatal cardiac surgery. The Laryngoscope. 2012;122(12):2781–5.

5. Al-Hakami H, Al Garni M. Incidence and risk factors for recurrent laryngeal nerve injury after thyroid surgery. Egypt J Otolaryngol. 2024;40(1):92.

6. Abu-Gameh AA, Koch JEJ, Schleifer D, Baruch Y, Engel I, Yaacobi E, et al. Incidence, risk factors, and outcomes of recurrent laryngeal nerve injury and dysphonia following anterior cervical spine surgery: a systematic review and meta-analysis. Cureus. 2025;17(2).

7. Chen YY, Chia YY, Wang PC, Lin HY, Tsai CL, Hou SM. Vocal cord paralysis and laryngeal trauma in cardiac surgery. Acta Cardiol Sin. 2017;33(6):624.

8. Liang KY, Scharpf J. Avoiding complications of thyroidectomy: recurrent laryngeal nerve and superior laryngeal nerve preservation. Otolaryngol Clin North Am. 2023;S0030-6665.

9. Xu W, Teng C, Ding G, Zhao N. Mechanisms of recurrent laryngeal nerve injury in endoscopic thyroidectomy for papillary thyroid carcinoma: A large data from C hina. Laryngoscope Investig Otolaryngol. 2023;8(2):604–9.

10. Wang H, Wang H, Li X, Xu W. Characteristics of Early Internal Laryngeal Muscle Atrophy After Recurrent Laryngeal Nerve Injurie s in Rats. The Laryngoscope. 2021;131(4):E1256–64.

11. Kumai Y, Ito T, Matsukawa A, Yumoto E. Effects of denervation on neuromuscular junctions in the thyroarytenoid muscle. The Laryngoscope. 2005;115(10):1869–72.

12. Goto T, Ueha R, Sato T, Fujimaki Y, Nito T, Yamasoba T. Single, high-dose local injection of bFGF improves thyroarytenoid muscle atrophy after paralysis. The Laryngoscope. 2020;130(1):159–65.

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13. Kodama H, Kumai Y, Nishimoto K, Sanuki T, Yumoto E. Modulation of satellite cells activity and M yo D in rat thyroarytenoid muscle after reinnervation. The Laryngoscope. 2015;125(7):E245–51.

14. McMullen CA, Andrade FH. Contractile dysfunction and altered metabolic profile of the aging rat thyroarytenoid muscle. J Appl Physiol. 2006;100(2):602–8.

15. Miyamaru S, Kumai Y, Ito T, Yumoto E. Effects of long-term denervation on the rat thyroarytenoid muscle. The Laryngoscope. 2008;118(7):1318–23.

16. Aires MM, de Oliveira KV, do Amaral JB, Mônico-Neto M, Martins JRM, Santiago AL, et al. Effect of testosterone on the thyroarytenoid muscle and lamina propria of female rat vocal folds. The Laryngoscope. 2024;134(5):2316–21.

17. Lenell C, Johnson AM. Sexual dimorphism in laryngeal muscle fibers and ultrasonic vocalizations in the adult rat. The Laryngoscope. 2017;127(8):E270–6.

18. Happak W, Zrunek M, Pechmann U, Streinzer W. Comparative histochemistry of human and sheep laryngeal muscles. Acta Otolaryngol (Stockh). 1989;107(3–4):283–8.

19. DelGaudio JM, Carroll WR, Sciote JJ, Esclamado RM. Atypical myosin heavy chain in rat laryngeal muscle. Ann Otol Rhinol Laryngol. 1995;104(3):237–45.

20. Schiaffino S, Reggiani C. Fiber types in mammalian skeletal muscles. Physiol Rev. 2011;91(4):1447–531.

21. DeNardi C, Ausoni S, Moretti P, Gorza L, Velleca M, Buckingham M, et al. Type 2X-myosin heavy chain is coded by a muscle fiber type-specific and developmentally regulated gene. J Cell Biol. 1993;123(4):823–35.

22. Aagaard P, Andersen JL. Correlation between contractile strength and myosin heavy chain isoform composition in human skeletal muscle. Med Sci Sports Exerc. 1998;30(8):1217–22.

23. Dedkov EI, Borisov AB, Carlson BM. Dynamics of postdenervation atrophy of young and old skeletal muscles: differential responses of fiber types and muscle types. J Gerontol A Biol Sci Med Sci. 2003;58(11):B984–91.

24. Wang Y, Pessin JE. Mechanisms for fiber-type specificity of skeletal muscle atrophy. Curr Opin Clin Nutr Metab Care. 2013 May;16(3):243–50. doi:10.1097/MCO.0b013e328360272d

25. Kostrominova TY, Dow DE, Dennis RG, Miller RA, Faulkner JA. Comparison of gene expression of 2-mo denervated, 2-mo stimulated-denervated, and control rat skeletal muscles. Physiol Genomics. 2005;22(2):227–43.

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26. Bakou S, Cherel Y, Gabinaud B, Guigand L, Wyers M. Type-specific changes in fibre size and satellite cell activation following muscle denervation in two strains of turkey (Meleagris gallopavo). J Anat. 1996;188(Pt 3):677.

27. Schiaffino S, Reggiani C. Fiber types in mammalian skeletal muscles. Physiol Rev. 2011.

28. Schiaffino S, Hughes SM, Murgia M, Reggiani C. MYH13, a superfast myosin expressed in extraocular, laryngeal and syringeal muscles. J Physiol. 2024;602(3):427–43.

29. Shiotani A, Flint PW. Myosin heavy chain composition in rat laryngeal muscles after denervation. The Laryngoscope. 1998;108(8):1225–9.

30. Phillips EG, Beggs LA, Ye F, Conover CF, Beck DT, Otzel DM, et al. Effects of pharmacologic sclerostin inhibition or testosterone administration on soleus muscle atrophy in rodents after spinal cord injury. PLoS One. 2018;13(3):e0194440.

31. Eng CM, Smallwood LH, Rainiero MP, Lahey M, Ward SR, Lieber RL. Scaling of muscle architecture and fiber types in the rat hindlimb. J Exp Biol. 2008;211(14):2336–45.

32. Bankhead P, Loughrey MB, Fernández JA, Dombrowski Y, McArt DG, Dunne PD, et al. QuPath: Open source software for digital pathology image analysis. Sci Rep. 2017;7(1):1–7.

33. Sawano S, Komiya Y, Ichitsubo R, Ohkawa Y, Nakamura M, Tatsumi R, et al. A one-step immunostaining method to visualize rodent muscle fiber type within a single specimen. PLoS One. 2016;11(11):e0166080.

34. Smith LR, Barton ER. SMASH–semi-automatic muscle analysis using segmentation of histology: a MATLAB application. Skelet Muscle. 2014;4(1):21.

35. Gartling G, Nakamura R, Bing R, Branski RC. A Novel Method for Thyroarytenoid Myofiber Culture. The Laryngoscope. 2023;133(11):3109–15.

36. Martin M. Cutadapt removes adapter sequences from high-throughput sequencing reads. EMBnet J. 2011;17(1):10–2.

37. Simon Andrews F, Segonds-Pichon A, Biggins L, Krueger C, Wingett S. FastQC: a quality control tool for high throughput sequence data. 2010.

38. Dobin A, Davis CA, Schlesinger F, Drenkow J, Zaleski C, Jha S, et al. STAR: ultrafast universal RNA-seq aligner. Bioinformatics. 2013;29(1):15–21.

39. Anders S, Pyl PT, Huber W. HTSeq—a Python framework to work with high-throughput sequencing data. bioinformatics. 2015;31(2):166–9.

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40. Love MI, Huber W, Anders S. Moderated estimation of fold change and dispersion for RNA-seq data with DESeq2. Genome Biol. 2014;15(12):550.

41. Subramanian A, Tamayo P, Mootha VK, Mukherjee S, Ebert BL, Gillette MA, et al. Gene set enrichment analysis: a knowledge-based approach for interpreting genome-wide expression profiles. Proc Natl Acad Sci. 2005;102(43):15545–50.

42. Huang R, Grishagin I, Wang Y, Zhao T, Greene J, Obenauer JC, et al. The NCATS BioPlanet–an integrated platform for exploring the universe of cellular signaling pathways for toxicology, systems biology, and chemical genomics. Front Pharmacol. 2019;10:445.

43. Zhou Y, Zhou B, Pache L, Chang M, Khodabakhshi AH, Tanaseichuk O, et al. Metascape provides a biologist-oriented resource for the analysis of systems-level datasets. Nat Commun. 2019;10(1):1523.

44. Leask A, Abraham DJ. TGF-β signaling and the fibrotic response. FASEB J. 2004;18(7):816–27.

45. Biernacka A, Dobaczewski M, Frangogiannis NG. TGF-β signaling in fibrosis. Growth Factors. 2011;29(5):196–202.

46. Delaney K, Kasprzycka P, Ciemerych MA, Zimowska M. The role of TGF-β1 during skeletal muscle regeneration. Cell Biol Int. 2017;41(7):706–15.

47. Miki Y, Morioka T, Shioi A, Fujimoto K, Sakura T, Uedono H, et al. Oncostatin M induces C2C12 myotube atrophy by modulating muscle differentiation and degradation. Biochem Biophys Res Commun. 2019;516(3):951–6.

48. Xiao F, Wang H, Fu X, Li Y, Ma K, Sun L, et al. Oncostatin M inhibits myoblast differentiation and regulates muscle regeneration. Cell Res. 2011;21(2):350–64.

49. Memme JM, Oliveira AN, Hood DA. p53 regulates skeletal muscle mitophagy and mitochondrial quality control following denervation-induced muscle disuse. J Biol Chem. 2022;298(2).

50. Xiang Y, You Z, Huang X, Dai J, Zhang J, Nie S, et al. Oxidative stress-induced premature senescence and aggravated denervated skeletal muscular atrophy by regulating progerin–p53 interaction. Skelet Muscle. 2022;12(1):19.

51. Fox DK, Ebert SM, Bongers KS, Dyle MC, Bullard SA, Dierdorff JM, et al. p53 and ATF4 mediate distinct and additive pathways to skeletal muscle atrophy during limb immobilization. Am J Physiol-Endocrinol Metab. 2014;307(3):E245–61.

52. Kjøbsted R, Hingst JR, Fentz J, Foretz M, Sanz MN, Pehmøller C, et al. AMPK in skeletal muscle function and metabolism. FASEB J. 2018;32(4):1741.

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53. Chen X, Li M, Chen B, Wang W, Zhang L, Ji Y, et al. Transcriptome sequencing and analysis reveals the molecular mechanism of skeletal muscle atrophy induced by denervation. Ann Transl Med. 2021;9(8):697.

54. Wagatsuma A, Osawa T. Time course of changes in angiogenesis-related factors in denervated muscle. Acta Physiol. 2006;187(4):503–9.

55. Ohn JH, Han SK, Park DJ, Park KS, Park YJ. Expression of thyroid stimulating hormone receptor mRNA in mouse C2C12 skeletal muscle cells. Endocrinol Metab. 2013;28(2):119.

56. Andreose J, Fumagalli G, Lømo T. Number of junctional acetylcholine receptors: control by neural and muscular influences in the rat. J Physiol. 1995;483(2):397–406.

57. Yumoto E. Pathophysiology and surgical treatment of unilateral vocal fold paralysis. Jpn Springer. 2015;6.

58. Dewan K, Vahabzadeh-Hagh A, Soofer D, Chhetri DK. Neuromuscular compensation mechanisms in vocal fold paralysis and paresis. The Laryngoscope. 2017;127(7):1633–8.

59. Shinners MJ, Goding GS, McLoon LK. Effect of recurrent laryngeal nerve section on the laryngeal muscles of adult rabbits. Otolaryngol Neck Surg. 2006;134(3):413–8.

60. Henry BM, Pękala PA, Sanna B, Vikse J, Sanna S, Saganiak K, et al. The anastomoses of the recurrent laryngeal nerve in the larynx: a meta-analysis and systematic review. J Voice. 2017;31(4):495–503.

61. Tracicaru RV, Bräuer L, Döllinger M, Tamba BI, Szilagyi A, Catalin CC, et al. Coactivation of the Laryngeal Muscles in Pigs Without External Neural Control Indicates Existence of an Intrinsic Neuronal Network. J Voice. 2024.

62. Tracicaru RV, Bräuer L, Döllinger M, Hînganu D, Paulsen F, Hînganu MV. Muscular and neuronal control of voice production–forgotten findings, current concepts, and new developments. Ann Anat-Anat Anz. 2024;255:152283.

63. Tracicaru RV, Bräuer L, Döllinger M, Schicht M, Tillmann B, Hînganu D, et al. Morphological Evidence for a Unique Neuromuscular Functional Unit of the Human Vocalis Muscle. Int J Mol Sci. 2024;25(22):11916.

64. Qiu X, Chen D, Li M, Gao Y, Liu F, Zheng H, et al. Transition of myosin heavy chain isoforms in human laryngeal abductors following denervation. Eur Arch Otorhinolaryngol. 2015;272(10):2915–23.

65. Zhang M, Chen M, Li Y, Rao M, Wang D, Wang Z, et al. Delayed denervation-induced muscle atrophy in Opg knockout mice. Front Physiol. 2023;14:1127474.

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66. Horton MJ, Rosen C, Close JM, Sciote JJ. Quantification of myosin heavy chain RNA in human laryngeal muscles: differential expression in the vertical and horizontal posterior cricoarytenoid and thyroarytenoid. The Laryngoscope. 2008;118(3):472–7.

67. Kemfack AM, Hernandez-Morato I, Moayedi Y, Pitman MJ. Transcriptome analysis of left versus right intrinsic laryngeal muscles associated with innervation. The Laryngoscope. 2024;134(8):3741–53.

68. Halasz G, Schmahl J, Negron N, Ni M, Lim WK, Atwal GS, et al. Optimized murine sample sizes for RNA sequencing studies revealed from large scale comparative analysis. Nat Commun. 2025;16(1):10173.

69. Hinrichsen C, Ryan A. The size of motor units in laryngeal muscles of the rat. Experientia. 1982;38(3):360–1.

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Table 1. Primary and secondary antibodies used in this study. Target antigen, host species, antigen species, isotype, product/source, and working dilution for each antibody. Primary antibodies were used to detect specific MyHC isoforms and laminin.

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Corresponding secondary antibodies were species- and isotype-specific and conjugated for detection.

Supplementary Table 1. Final morphometric biological replicate numbers by muscle, fiber type, and sex. Final n values reflect tissue quality, section availability, and fiber-type-specific analyzability after segmentation and quality control. Male and female counts indicate the number of animals contributing to each morphometric comparison. Putative IIX/IIX-like fibers were operationally classified based on absence of BF-35 immunoreactivity and should not be interpreted as definitive pure IIX fibers. Abbreviations: Gastroc, gastrocnemius; LTA, lateral thyroarytenoid; MTA, medial thyroarytenoid; MyHC, myosin heavy chain; PCA, posterior cricoarytenoid.

Supplementary Table 2. RNA-seq sample quality metrics. Sample-level RNA integrity and sequencing quality metrics are reported for each RNA-seq sample. Metrics include RNA integrity number, sequencing quality, alignment rate, detected genes, and reads retained in count matrices. Abbreviations: RIN, RNA integrity number; RNA-seq, RNA sequencing; Q30, percentage of bases with Phred quality score ≥30.

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Supplementary Table 3. KEGG/Enrichr pathway enrichment at the ±4-fold-change threshold. The table reports exported KEGG/Enrichr enrichment results generated from tissue-specific ±4-fold-change DEG lists. Upregulated and downregulated gene lists were analyzed separately for each muscle. Columns include muscle, pathway term, Enrichr p-value, Enrichr q-value, and direction of regulation. Abbreviations: DEG, differentially expressed gene; Enrichr q-value, multiple-testing-adjusted enrichment value; Gastroc, gastrocnemius; LTA, lateral thyroarytenoid; MTA, medial thyroarytenoid; PCA, posterior cricoarytenoid.

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Supplementary Table 4. Pairwise comparisons of denervation-induced percent change in minimum Feret diameter across muscle/fiber-type groups. Pairwise comparisons were performed using one-way ANOVA with Tukey's post hoc test. Percent change was calculated for each animal as the denervated side relative to the contralateral internal-reference side. Mean differences are reported in percentage points. Positive mean differences indicate that the first listed group had a higher, less negative percent-change value than the second listed group, consistent with relatively less atrophy. Adjusted p values are Tukey-corrected for multiple comparisons. IIX fibers were operationally classified as putative IIX/IIX-like fibers based on absence of BF-35 immunoreactivity and should not be interpreted as definitive pure IIX fibers. Abbreviations: CI, confidence interval; Gastroc, gastrocnemius; LTA, lateral thyroarytenoid; MTA, medial thyroarytenoid; MyHC, myosin heavy chain; mFD, minimum Feret diameter; ns, not significant; PCA, posterior cricoarytenoid. Significance: ns, adjusted $p > 0.05$; * adjusted $p \leq 0.05$; ** adjusted $p \leq 0.01$; *** adjusted $p \leq 0.001$; **** adjusted $p \leq 0.0001$.

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Supplementary Table 5. Tissue-specific differentially expressed genes at the ±4-fold-change threshold. Upregulated and downregulated genes are listed by tissue and direction of regulation. Genes available from the pathway/gene-block output are color-coded in the relevant tissue columns to facilitate interpretation of pathway-associated gene patterns. Color coding is visual only and does not alter the DEG results. Abbreviations: DEG, differentially expressed gene; Gastroc, gastrocnemius; LTA, lateral thyroarytenoid; MTA, medial thyroarytenoid; PCA, posterior cricoarytenoid.

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Supplementary Table 6. BioPlanet-enriched pathways at the ±1.5-fold-change threshold. The table reports BioPlanet pathway enrichment results generated from tissue-specific gene lists at the exploratory ±1.5-fold-change threshold. Pathways are organized by tissue and direction of regulation where applicable. These results were used as exploratory pathway-level context and should be interpreted as supportive rather than definitive mechanistic evidence. Abbreviations: BioPlanet, NCATS BioPlanet pathway database; FDR, false discovery rate; Gastroc, gastrocnemius; LTA, lateral thyroarytenoid; MTA, medial thyroarytenoid; PCA, posterior cricoarytenoid.

Supplementary Figure 1. Volcano plots showing differential expression across tissue comparisons. Volcano plots show differential expression for (A) gastrocnemius, (B) soleus, (C) medial thyroarytenoid (MTA), (D) lateral thyroarytenoid (LTA), and (E) posterior cricoarytenoid (PCA). Each point represents one detected gene. The x-axis shows log2 fold-change for denervated/injured tissue relative to contralateral internal reference tissue, and the y-axis shows −log10 adjusted p-value. Vertical lines indicate ±1.5-fold change (log2FC = ±0.585) and ±4-fold change (log2FC = ±2). Purple points indicate downregulated genes and green points indicate upregulated genes. Dark purple/green indicates genes meeting the ±1.5-fold threshold but not the ±4-fold threshold; light purple/green indicates genes meeting or exceeding the ±4-fold threshold. Gray points indicate genes not meeting fold-change and/or adjusted p-value criteria. The ±1.5-fold thresholds are shown for visualization of broader expression patterns, whereas the ±4-fold threshold was used for the primary pathway-enrichment analyses.

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Supplementary Figure 2. KEGG/Enrichr pathway enrichment summary at the ±4-fold-change threshold. Dot plot summarizing KEGG/Enrichr pathway enrichment using tissue-specific ±4-fold-change DEG lists. Upregulated and downregulated gene lists were analyzed separately for each muscle. Dot color indicates gene-list direction, with green representing upregulated lists and purple representing downregulated lists. Dot size reflects −log10(Enrichr q-value), and black outlines indicate terms meeting the primary FDR threshold of Enrichr q ≤ 0.05. Terms with Enrichr q-values >0.05 and ≤0.10 are included only as exploratory visual context and were not interpreted as statistically significant.

Figure 1. Morphologic and quantitative assessment of denervation-induced atrophy in laryngeal and hindlimb muscles. (A) Representative immunofluorescent images of muscle sections from the denervated side and contralateral internal-reference side for MTA, LTA, PCA, soleus, and gastrocnemius. Myosin heavy chain isoforms were labeled to identify fiber types: fibers positive for “all-but-IIX,” IIA, or MyHC I appear green; IIB fibers appear orange. Laminin outlines fiber boundaries in blue. Post-analysis brightness adjustments were presentation-only and did not affect segmentation or quantitative results. Scale bars: MTA and LTA = 20 µm; PCA = 50 µm; soleus and gastrocnemius = 100 µm. (B) Scatter plot showing percent change in mean minimum Feret diameter (mFD) for each animal and fiber type, calculated as denervated side relative to contralateral internal-reference side. Negative values indicate atrophy. Horizontal lines indicate the median. Statistical significance was determined by one-way ANOVA with Tukey’s post hoc test. Laryngeal muscles exhibited significantly less atrophy than fiber-matched hindlimb muscles, including MTA putative IIX/IIX-like versus

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gastrocnemius putative IIX/IIX-like, PCA IIA versus soleus IIA, and LTA IIB versus gastrocnemius IIB. Greater atrophy was observed in LTA IIB fibers compared with MTA putative IIX/IIX-like fibers.

Figure 2. Pathway enrichment in gastrocnemius. Heatmaps show log2-normalized gene expression after subtracting the per-gene mean, with orange indicating increased expression and blue indicating decreased expression. Horizontal bars at the top of each heatmap indicate sample groups. At a fold-change threshold of ±4, significant enrichment was observed in (A) Oncostatin M signaling, (B) cytokine–cytokine receptor interaction, (C) TGF-β regulation of extracellular matrix, and (D) leptin influence on immune response.

Figure 3. TP53-associated enrichment in PCA tissue. Heatmap displays log2-normalized gene expression after subtracting the per-gene mean, with orange indicating increased expression and blue indicating decreased expression. Horizontal bars at the top indicate sample groups. At a fold-change threshold of ±4, the TP53 network was significantly enriched in PCA tissue. The TP53-associated enrichment signal was driven by a small set of upregulated genes shown in the heatmap, including Cdkn1a, Tp63, Bid, and Gadd45a. This heatmap should be interpreted as a gene-set enrichment summary rather than a complete canonical TP53 pathway map.

Figure 4. Gene Ontology enrichment across hindlimb and laryngeal muscles. Heatmaps show significantly enriched Gene Ontology terms identified from tissue-

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specific ±4-fold-change gene lists using Metascape. Columns correspond to individual muscle-specific gene lists from hindlimb muscles and intrinsic laryngeal muscles, including gastrocnemius, soleus, MTA, LTA, and PCA. Rows represent enriched GO terms. Color intensity reflects enrichment significance expressed as −log10 p-value, with darker colors indicating stronger enrichment. (A) GO terms enriched among upregulated gene lists. (B) GO terms enriched among downregulated gene lists. This analysis highlights conserved denervation-associated processes as well as organ-selective differences between hindlimb and laryngeal muscles.

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